

1. COMPLETE BILLABLE TECHNOLOGY INVENTORY (ALL PAID EXTERNAL VENDORS & APIS)

Tech / Service	Module / Role	Billing Metric & Pricing	Free Tier Allowance	Normal Expected Usage	Worst-Case Stress Behavior	Safeguard in Code
Google Gemini LLM gemini-3.1-flash-lite	AI Stylist Pipeline (Intent, Compose, RAG)	In: \$0.075 / 1M tokens Out: \$0.30 / 1M tokens	First 15 RPM / free trial tier	~2.3k in / ~580 out tokens ~\$0.0020 USD (0.100 EGP)/turn	Max daily messages burned every single day across all tiers	Atomic CAS daily limits via UsageCounter; 429 quota block
Google Gemini Vision gemini-3.1-flash-lite	Wardrobe Item Classifier (BullMQ)	\$0.075 / 1M tokens (~258 tokens / image)	Included in Gemini API tier	408 in / 200 out tokens \$0.0007 USD (0.035 EGP)/item	User uploads 100% max wardrobe items instantly	Runs strictly once per item; cached forever in MongoDB
Virtual Try-On Nano Banana Models (Gemini Flash Image Lite)	Phase 15F Try-On (Nano Banana 2 Lite)	\$0.0336 USD / generated image (1.680 EGP / image)	None (Paid tiers only)	0 on Free (Disabled); 4 to 10 try-ons / month on paid	User consumes 100% monthly try-on cap	Billed before queue job; deterministic deduplication
Serper.dev (Shopping) Google Shopping Egypt	Product Search & Wardrobe Gaps (15E)	\$50.00 / 50,000 queries (~2,500 EGP) \$0.0010 USD (0.050 EGP)/query	2,500 queries free on signup	1 to 3 queries per gap; 50% absorbed by Redis cache	0% cache hit, user burns 100% monthly search quotas	24-hour Redis Cache ; monthly hard cap (15–60/mo)
MongoDB Atlas M10 Replica Set	Core Database & ACID Transactions	~\$57.00 – \$60.00 USD / mo (~2,850 – 3,000 EGP / mo)	M0 Shared (Dev/testing only)	Standard operational DB reads, writes, and aggregations	Burst IOPS from mass concurrent users	M10 handles 1,500 IOPS; replica set required for Phase 5 transactions
Upstash Vector Serverless Vector DB	Garment Retrieval & Editorial RAG	\$0.20 / 100k queries \$0.25 / 100k vectors / mo	10,000 queries / day 100,000 vector records	Free tier completely covers 5,000 active users	Burst search traffic past 10k daily requests (\$0.20/100k)	Namespaced per user; cached editorial knowledge vectors
Upstash / Cloud Redis In-Memory Store	BullMQ Queues & 24h Search Cache	\$0.20 / 100k commands or ~\$15.00 USD / mo dedicated	10,000 commands / day	BullMQ job coordination & 24h search cache key reads	Heavy BullMQ job retries & cache lookups	Job attempts bounded to 2 with exponential backoff
Cloudinary Media CDN & Storage	Wardrobe, Shape Models & KYC Images	Free (25 credits) $\rightarrow$ \$89.00 USD / mo (~4,450 EGP / mo)	25 Monthly Credits (~25k transformations / 25GB)	Free tier covers early stage; Plus plan covers 5,000 users	Users upload full wardrobe quota + uncompressed photos	Sharp in-memory image compression before upload
Resend Transactional Email	Auth OTP, Booking, Receipts	Free (3k/mo) $\rightarrow$ \$20.00 USD / mo (~1,000 EGP / mo)	3,000 emails / month (100 emails / day)	OTP verification & booking confirmation emails	Account creation spam & login OTP abuse	OTP rate-limiting (3 attempts / 15 mins) & Redis cooldown
Firebase (Firestore/FCM) Google Cloud Platform	Real-Time Chat & Push Notifications	Firestore: \$0.06/100k reads FCM: 100% Free forever	50k reads / 20k writes daily (Spark Free Plan)	Active booking chat messages and mobile push notifications	Massive continuous chat messaging in active bookings	Chat locked upon session completion; read-only archive
Paymob Payment Gateway Egyptian Merchant Engine	Cards, Meeza & Mobile Wallets	2.75% + 3.00 EGP per card transaction	0 EGP monthly maintenance	Transaction fee deducted automatically on paid checkout	High chargeback or refund rate	Double-entry Ledger piastres reconciliation; anti-fraud CAS
Server Hosting (Compute) DigitalOcean / Render	API Server + BullMQ Worker Runner	~\$24.00 USD / mo (4GB RAM) (~1,200 EGP / mo)	Free tier on hobby plans	Runs Node.js API process + background worker daemon	CPU spikes during concurrent LLM streaming	Stateless containers; horizontal autoscale capability

2. PLAN-BY-PLAN WORST-CASE ECONOMICS (100% QUOTA BURN + GATEWAY FEES)

User Plan & Monthly Price	Max AI Chat Spend (100% Daily Ceiling)	Wardrobe Upload (100% Capacity)	Product Search (Max Serper Calls)	Virtual Try-On (100% Quota)	Paymob Fee (2.75% + 3 EGP)	Total Variable Cost (EGP)	Worst-Case Net Margin
Client Free 0 EGP (\$0.00)	1.00 EGP (10 lifetime)	0.25 EGP (7 items)	0.00 EGP (0 searches)	0.00 EGP (0 try-ons)	0.00 EGP	1.25 EGP (\$0.025)	CAC: -1.25 EGP
Client Basic 50 EGP (\$1.00)	15.00 EGP (150 msgs)	0.88 EGP (25 items)	2.25 EGP (15 x 3 calls)	6.72 EGP (4 try-ons)	4.38 EGP	29.23 EGP (\$0.585)	+20.77 EGP (+41.5%)

User Plan & Monthly Price	Max AI Chat Spend (100% Daily Ceiling)	Wardrobe Upload (100% Capacity)	Product Search (Max Serper Calls)	Virtual Try-On (100% Quota)	Paymob Fee (2.75% + 3 EGP)	Total Variable Cost (EGP)	Worst-Case Net Margin
Client Mid 150 EGP (\$3.00)	105.00 EGP (1,050 msgs)	1.58 EGP (45 items)	4.50 EGP (30 x 3 calls)	10.08 EGP (6 try-ons)	7.13 EGP	128.29 EGP (\$2.566)	+21.71 EGP (+14.5%)
Client Pro 250 EGP (\$5.00)	240.00 EGP (2,400 msgs)	3.50 EGP (100 items)	6.75 EGP (45 x 3 calls)	13.44 EGP (8 try-ons)	9.88 EGP	273.57 EGP (\$5.471)	-23.57 EGP (-9.4%)*
Client Enterprise 500 EGP (\$10.00)	450.00 EGP (4,500 msgs)	7.00 EGP (200 items)	9.00 EGP (60 x 3 calls)	16.80 EGP (10 try-ons)	16.75 EGP	499.55 EGP (\$9.991)	+0.45 EGP (+0.1%)
Stylist Plans (All) 50 / 125 / 250 EGP	0.00 EGP (No AI)	0.00 EGP	0.00 EGP	0.00 EGP	4.38 to 9.88 EGP	4.38 – 9.88 EGP	+91.2% to +96.0%

* Note on Pro Plan worst case: A user must send 80 full styling prompts every single day of the month without missing a single day. In actual usage, users average 15–20% of their daily cap, yielding +35% to +45% healthy profit.

3. 5,000 ACTIVE USERS COHORT SIMULATION IN ABSOLUTE WORST CASE

Financial & Operational Line Item	Realistic User Consumption (Average)	Theoretical Worst Case (100% Cap Burn)	Operational Commentary & Safety Analysis
Client Subscription Revenue	152,500 EGP (\$3,050 USD)	152,500 EGP (\$3,050 USD)	1,000 Basic + 350 Mid + 100 Pro + 50 Enterprise paid clients
Stylist Subscription Revenue	21,250 EGP (\$425 USD)	21,250 EGP (\$425 USD)	200 paid marketplace stylists (100 Basic, 70 Pro, 30 Enterprise)
TOTAL MONTHLY REVENUE	173,750 EGP (\$3,475 USD)	173,750 EGP (\$3,475 USD)	Steady monthly MRR from recurring active subscriptions
Total AI Variable Costs (Gemini + Serper + TryOn)	85,510 EGP (\$1,710 USD)	117,760 EGP (\$2,355 USD)	Worst case assumes 100% maximum allotment burned
Paymob Payment Processing Fees	10,464 EGP (\$209 USD)	10,464 EGP (\$209 USD)	2.75% + 3 EGP per checkout transaction across all tiers
Fixed Tech Stack (Atlas M10 + Droplet + Cloudinary + Upstash + Resend)	10,650 EGP (\$213 USD)	10,650 EGP (\$213 USD)	Fixed monthly infrastructure retainers to run production
TOTAL MONTHLY EXPENSES	106,624 EGP (\$2,132 USD)	138,874 EGP (\$2,777 USD)	Includes all 11 external vendor services combined
NET OPERATING PROFIT	+67,126 EGP (\$1,343 USD)	+34,876 EGP (\$698 USD)	REMAINS NET PROFITABLE EVEN IN WORST CASE!
BLENDED OPERATING MARGIN	+38.6%	+20.1%	Business survives and generates cash flow under maximum stress

4. FOUR CORE ARCHITECTURAL CIRCUIT BREAKERS PROTECTING CAPITAL

1. Atomic CAS Daily Quotas All message limits are metered by atomic Compare-And-Swap (\$inc with used <= limit - count) in Cairo business timezone. Prevents race-condition spikes and bounds daily financial liability to small fractions of a dollar.	2. In-Memory Image Pre-Compression Before any photo reaches Cloudinary or Gemini Vision, the upload service compresses images in-memory via Sharp . This shrinks payload sizes by 75–85%, keeping Cloudinary credit burn safely inside standard plan limits.	3. 24h Redis Shopping Cache Common wardrobe gaps (e.g. "navy formal trousers", "white sneakers") are stored in Redis for 24 hours. When multiple users ask for similar occasions, Serper API queries are bypassed, slashing search costs by 50–70%.	4. Google Hard Billing Ceiling By establishing a monthly hard budget cutoff (e.g. \$2,000 USD) in Google Cloud Console, the Gemini API automatically refuses excess calls before unexpected cloud debt can ever accumulate.
---	---	---	---